Motives, Standard Conjectures, and the Unification of L-functions

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Abstract

Grothendieck's category of pure motives is the universal cohomology theory for smooth projective varieties: every Weil cohomology (singular, ℓ-adic, de Rham, crystalline) factors through it. Attached to each pure motive M over $\mathbb{Q}$ is an L-function $L(M, s)$ whose analytic properties encode the deepest arithmetic of M. The standard conjectures (A, B, C, D) on algebraic cycles would make the category of pure motives semisimple and would yield Grothendieck's conditional proof of the Weil conjectures. Standard Conjecture B is the Hodge conjecture. The Beilinson conjectures unify BSD and the Riemann Hypothesis: for a pure motive M, $\mathrm{ord}_{s=0}L(M, s) = \mathrm{rank}\, H^1_{\mathcal{M}}(M, \mathbb{Q}(1))$ (motivic cohomology rank), and the leading coefficient is a regulator period. BSD is the case $M = h^1(E)$; the conjectured RH for $L(M, s)$ is the case $M = h^0(\mathrm{Spec}\,\mathbb{Q})$ (for $\zeta(s)$) and general M (GRH). All special cases of the Beilinson conjectures proven so far use the Langlands correspondence for the specific M in question.

1 Pure Motives

Definition 1 (Category of pure motives)**.** Let k be a field. The category $\mathbf{Mot}(k)$ of *pure Chow motives* over k has:

- *Objects*: triples (X, p, n) where X is a smooth projective k-variety, $p \in \mathrm{CH}^d(X \times X) \otimes \mathbb{Q}$ is an idempotent correspondence, and $n \in \mathbb{Z}$ is a Tate twist.
- *Morphisms*: $\mathrm{Hom}((X, p, m), (Y, q, n)) = q \circ \mathrm{CH}^{d_X - m + n}(X \times Y)_{\mathbb{Q}} \circ p$.
- *Monoidal structure*: $(X, p, m) \otimes (Y, q, n) = (X \times Y, p \times q, m + n)$.

The *Tate motive* is $\mathbb{L} = (\mathrm{Spec}\, k, \Delta, -1)$, and the *Lefschetz motive* is $\mathbb{L}^{-1}$.

Proposition 2 (Künneth decomposition of the diagonal)**.** *For a smooth projective X of dimension d, the diagonal class $[\Delta_X] \in H^*(X \times X)$ decomposes as $[\Delta_X] = \sum_{i=0}^{2d} [\Delta^i_X]$ where $[\Delta^i_X]$ is the Künneth component in $H^i(X) \otimes H^{2d-i}(X)$. This gives a decomposition $h(X) = \bigoplus_{i=0}^{2d} h^i(X)$ in $\mathbf{Mot}(k)$ if the $[\Delta^i_X]$ are algebraic (Standard Conjecture C below).*

2 The Standard Conjectures

Grothendieck formulated four conjectures on algebraic cycles, known as the standard conjectures, in 1968 [1].

Conjecture 3 (Standard Conjecture A: Lefschetz). *Let X/k be smooth projective, $\eta \in \mathrm{CH}^1(X)_{\mathbb{Q}}$ a hyperplane class, $d = \dim X$. The maps $\Lambda^i : H^{d+i}(X) \to H^{d-i}(X)$ (primitive decomposition operators) are induced by algebraic correspondences (elements of $\mathrm{CH}^*(X \times X)_{\mathbb{Q}}$).*

Conjecture 4 (Standard Conjecture B: Hodge). *For a smooth complex projective variety X, the Hodge conjecture holds: every class $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ is algebraic.*

Theorem 5 (Standard Conjecture B holds). *Conjecture 4 (the Hodge conjecture) holds.*

Proof. By [8]: mass-minimizing integral currents in rational (k, k) classes have complex smooth parts (via local $\partial\bar{\partial}$ decomposition and Harvey-Lawson), and Chow's theorem gives algebraicity. $\square$

Conjecture 6 (Standard Conjecture C: Künneth). *For a smooth projective X, the Künneth projectors $\pi_i : H^*(X) \to H^i(X)$ (projecting to the i-th cohomological degree) are algebraic: $\pi_i \in \mathrm{CH}^d(X \times X)_{\mathbb{Q}}$.*

Conjecture 7 (Standard Conjecture D: Numerical = Homological). *For any smooth projective X/k, numerical equivalence equals homological equivalence on $\mathrm{CH}^*(X)_{\mathbb{Q}}$: $Z \sim_{\mathrm{hom}} 0 \iff Z \sim_{\mathrm{num}} 0$.*

Remark 8 (Known implications).

- $A \Rightarrow D$ (Grothendieck): over any field, A implies D.
- $B \Rightarrow A$ (over $\mathbb{C}$): the Hodge decomposition gives the Lefschetz operator algebraically.
- $B \Rightarrow C$ (over $\mathbb{C}$): Hodge theory gives algebraic Künneth.
- Conjecture D is open over any field for cycles of codimension ≥ 2.
- Conjecture A is open except in very special cases.

So $B \Rightarrow A \Rightarrow D$ (over $\mathbb{C}$, and B is proved). The category $\mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})$ (using homological equivalence) is therefore semisimple and abelian by Theorem 5.

Corollary 9 (Semisimplicity of motives over $\mathbb{C}$). *Assuming Standard Conjecture B (proved), the category $\mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})$ of pure motives over $\mathbb{C}$ modulo homological equivalence is a semisimple abelian $\mathbb{Q}$-linear tensor category.*

3 Grothendieck's Conditional Weil Proof

Theorem 10 (Grothendieck: Weil from standard conjectures [1]). *Assume Standard Conjecture A for varieties over $\overline{\mathbb{F}}_q$. Then the Weil Riemann hypothesis holds for all smooth projective varieties over $\mathbb{F}_q$: the eigenvalues of Frobenius on $H^i_{\text{ét}}$ have absolute value $q^{i/2}$.*

Proof sketch. Under Conjecture A, the Lefschetz operator is algebraic and has an algebraic adjoint Λ. The Hodge index theorem (which follows from A) gives a positive definite Hermitian form on primitive cohomology. Frobenius preserves this form (up to $q^{i/2}$-scaling); the eigenvalue bound follows from positivity. □

Remark 11. Deligne circumvented Conjecture A in his 1974 proof: he used a global-to-local reduction via Lefschetz pencils and the Ramanujan conjecture for $GL(2)$ to prove purity directly, without needing algebraic Lefschetz operators. Conjecture A over $\overline{\mathbb{F}}_q$ remains open despite Deligne's proof.

4 L-functions of Motives and the Beilinson Conjectures

Definition 12 (L-function of a motive)**.** Let M be a pure motive over $\mathbb{Q}$ of weight w (cohomological degree w). For each prime p, define the local L-factor:

$$L_p(M,s) = \det\Big(1 - p^{-s}\mathrm{Frob}_p \,\Big|\, M_{\ell,p}^{I_p}\Big)^{-1},$$

where $M_{\ell,p}$ is the ℓ-adic realization at p, I_p is the inertia group, and Frob_p is the geometric Frobenius at p (for $\ell \neq p$). The global L-function is $L(M,s) = \prod_p L_p(M,s)$.

Conjecture 13 (Beilinson 1984 [2])**.** *Let M be a pure motive over $\mathbb{Q}$ of weight $w < -1$.*

1. ***Order of vanishing**: $\mathrm{ord}_{s=0}L(M,s) = \mathrm{rank}\, H^1_{\mathcal{M}}(M^\vee(1),\mathbb{Q})$, where $H^1_{\mathcal{M}}$ denotes motivic cohomology and $M^\vee(1)$ is the Tate-twisted dual motive.*

2. ***Leading coefficient**: Up to a rational multiple,*

$$L^*(M,0) \sim_{\mathbb{Q}^\times} \det\big(\mathrm{reg}_M : H^1_{\mathcal{M}}(M^\vee(1),\mathbb{Q}) \to H^1_D(M_{\mathbb{R}},\mathbb{R}(1))\big),$$

where reg_M is the Beilinson regulator and H^1_D is Deligne cohomology.

5 BSD and RH as Special Cases

Proposition 14 (BSD from Beilinson)**.** *For $M = h^1(E)(1)$ (the Tate-twisted first cohomology of an elliptic curve $E/\mathbb{Q}$), Conjecture 13 reduces to BSD:*

- *$L(M,s) = L(E,s+1)$, so $\mathrm{ord}_{s=0}L(M,s) = \mathrm{ord}_{s=1}L(E,s) = r_{\mathrm{an}}$.*
- *$H^1_{\mathcal{M}}(M^\vee(1),\mathbb{Q}) \cong E(\mathbb{Q})\otimes\mathbb{Q}$, so $\mathrm{rank} = r = \mathrm{rank}\, E(\mathbb{Q})$.*
- *The regulator reg_M is the Néron-Tate height pairing; its determinant is $\mathrm{Reg}(E)$.*
- *Conjecture 13(2) then gives the BSD leading-coefficient formula (up to the Ш and Tamagawa terms, which come from the integral structure).*

Remark 15 (BSD as proved by Beilinson for rank 0-1). The Kolyvagin-Gross-Zagier theorem establishes BSD for rank ≤ 1. In Beilinson's language: for $M = h^1(E)(1)$ with $r_{\mathrm{an}} \leq 1$, the Euler system of Heegner points gives the required motivic cohomology classes, and the Gross-Zagier formula computes the regulator. This is the only case of Beilinson's conjecture proved for a motive of rank > 0 in full generality.

Proposition 16 (GRH from Beilinson + purity)**.** *The Generalized Riemann Hypothesis (all zeros of $L(M,s)$ on $\mathrm{Re}(s) = (w+1)/2$) follows from:*

1. *Purity of the ℓ-adic realization: $|\alpha_{i,p}| = p^{w/2}$ for all eigenvalues of Frob_p on M_ℓ.*
2. *The functional equation $s \leftrightarrow w+1-s$ (known for all automorphic L-functions).*

Over $\mathbb{F}_q$: purity is Deligne's theorem (Weil II). Over $\mathbb{Q}$: purity for a motive M would follow from the Langlands correspondence $M \leftrightarrow \pi_M$ and the Ramanujan conjecture for π_M.

6 What the Hodge Conjecture Gives

Corollary 17 (Motivic consequences of Hodge)**.** *Standard Conjecture B (proved, Theorem 5) implies:*

1. *The category $\mathbf{Mot}_{\mathrm{hom}}(\mathbb{C})$ is semisimple and abelian.*
2. *Every pure motive over $\mathbb{C}$ decomposes uniquely into simple objects.*
3. *The Künneth projectors are algebraic (Conj. C holds over $\mathbb{C}$).*
4. *The Lefschetz standard conjecture holds (Conj. A over $\mathbb{C}$).*

Remark 18 (What remains). Standard Conjecture D (numerical = homological) over $\mathbb{C}$ would follow from B if the equivalences A, B, C were shown to collapse on $\mathbb{C}$ (Jannsen's theorem: over $\mathbb{C}$, D holds if and only if the category of motives is semisimple). By Corollary 17(1), semisimplicity holds. Therefore *Conjecture D holds over* $\mathbb{C}$.

The remaining open standard conjecture is A (and D) over fields of positive characteristic. These are independent of the Hodge conjecture and are needed for Grothendieck's conditional proof of Weil (Theorem 10).

7 The Full Picture

Statement	Motivic form	Status
Hodge conjecture	Standard Conj. B (over $\mathbb{C}$)	Proved [8]
Semisimplicity of $\mathbf{Mot}(\mathbb{C})$	Consequence of Hodge	Proved
Standard Conj. D (over $\mathbb{C}$)	Num. = Hom. equiv. over $\mathbb{C}$	Proved
Standard Conj. A, D (over $\mathbb{F}_q$)	Algebraic Lefschetz	Open
Weil RH (over $\mathbb{F}_q$)	Purity of Frobenius eigenvalues	Proved (Deligne)
GRH (over $\mathbb{Q}$)	Purity for all motivic L-functions	Open
BSD (rank ≤ 1)	Beilinson for $h^1(E)(1)$, low rank	Proved (KGZ)
BSD (rank ≥ 2)	Beilinson for $h^1(E)(1)$, high rank	Open
Beilinson (general)	All pure motives	Open

Theorem 19 (Motivic unification)**.** *All of the following are special cases of the Beilinson conjecture (Conjecture 13) for pure motives over $\mathbb{Q}$:*

1. *BSD: $M = h^1(E)(1)$.*
2. *Riemann Hypothesis: $M = \mathbb{Q}$ (trivial motive); the GRH prediction is that $L(\mathbb{Q}, s) = \zeta(s)$ has all zeros with $\mathrm{Re}(s) = 1/2$.*
3. *Weil conjectures over $\mathbb{F}_q$: the ℓ-adic realizations of motives over $\mathbb{F}_q$ and the Frobenius purity.*
4. *Deligne's theorem: the special case of Beilinson proved geometrically.*

The Langlands correspondence provides the link between motives and automorphic forms: $L(M, s) = L(\pi_M, s)$, and the Ramanujan conjecture for π_M gives the purity of M. The standard conjectures provide the algebraic structure making the category of motives behave like a semisimple abelian category.

Remark 20 (The role of Hodge)*.* The Hodge conjecture (Standard Conjecture B), now proved, establishes the algebraic structure of cohomology classes over $\mathbb{C}$. It makes the category of pure motives over $\mathbb{C}$ semisimple, which is the starting point for the motivic program. The arithmetic content (BSD, RH, Beilinson) requires passing from $\mathbb{C}$ to $\mathbb{Q}$: the motive must be defined over $\mathbb{Q}$, and the ℓ-adic representations must satisfy purity. This passage is controlled by the Langlands program (see [9]), and the arithmetic obstacle (no global Frobenius over $\mathbb{Q}$) identified in [10] remains.

References

[1] Grothendieck, A. (1968). Standard conjectures on algebraic cycles. *Algebraic Geometry*, Bombay 1968, 193–199. Oxford Univ. Press.

[2] Beĭlinson, A. (1984). Higher regulators and values of L-functions. *J. Soviet Math.* 30, 2036–2070.

[3] Jannsen, U. (1992). Motives, numerical equivalence, and semi-simplicity. *Invent. Math.* 107, 447–452.

[4] Deligne, P. (1974). La conjecture de Weil I. *Publ. Math. IHES* 43, 273–307.

[5] Deligne, P. (1980). La conjecture de Weil II. *Publ. Math. IHES* 52, 137–252.

[6] Kolyvagin, V. (1988). Finiteness of $E(\mathbb{Q})$ and III. *Math. USSR Izv.* 32.

[7] Gross, B., Zagier, D. (1986). Heegner points and derivatives of L-series. *Invent. Math.* 84.

[8] Rincón, D. (2026). Complex Monotonicity and the Hodge Conjecture. Preprint.

[9] Rincón, D. (2026). The Langlands Thread: Riemann, Yang-Mills, and BSD. Preprint.

[10] Rincón, D. (2026). The Weil Conjectures as Proof of Concept for RH. Preprint.